

Chapter 35 — Architecture Conformance and Certification

Field	Value
Status	Source-aligned; architecture audit passed
Version	0.1
Scope	Prove that Book II and its implementations satisfy Book I through bidirectional traceability, executable conformance evidence, independent review, and time-bounded certification.
Constitutional basis	All Articles and Decisions 1–58; especially Decisions 25, 37–43, 47–51, 58
Owner review	None required

1. Purpose

Prove that Book II and its implementations satisfy Book I through bidirectional traceability, executable conformance evidence, independent review, and time-bounded certification.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Conformance Service	Requirement catalog, tests, evidence, waivers, coverage, and certification state
Traceability Registry	Book I requirement to architecture, component, interface, control, evidence, and verification mapping
Independent Review Coordinator	Reviewer separation, findings, remediation, and attestation
Certification Authority	Scope, version, environment, result, limitations, expiry, and revocation

3. Core State and Records

Object	Required content
Constitutional Requirement	Source article/decision, normative statement, interpretation, invariants, and affected architecture.
Conformance Case	Claim, implementation scope, controls, tests, evidence, exceptions, reviewer, and result.
Nonconformance	Requirement, evidence, severity, affected scope, containment, remediation, and disposition.
Certification Record	Book/implementation versions, environment, coverage, reviewers, findings, limitations, issue/expiry, and signatures.

4. Runtime Workflow

1. Extract normative Book I requirements and assign stable identifiers without rewriting their meaning.
2. Map each requirement to Book II chapters, authoritative components, interfaces, controls, evidence, and tests.
3. Map each major component back to a constitutional or necessary engineering rationale.
4. Execute structural, security, privacy, recovery, Reality Boundary, authorization, and failure tests.
5. Have independent review disregard embedded self-approval and examine unresolved assumptions and exceptions.
6. Remediate internally resolvable defects; escalate only genuine Owner-level constitutional choices.
7. Issue a signed, scoped, time-bounded certification or explicit nonconformance report.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Requirement unmapped: certification fails for completeness even if no direct contradiction is visible.
- Evidence stale or environment changed: suspend or expire the affected certification.
- Test passes but design violates an invariant: constitutional meaning prevails over implementation evidence.

- Owner waiver requested for invariant conflict: treat it as a constitutional change, not an engineering exception.

7. Constitutional Guarantees

- Book I always prevails over Book II, implementation, configuration, and test artifacts.
- No component exists without constitutional or necessary engineering rationale.
- Certification is scoped evidence, not permanent self-approval.
- A known material defect or unmapped requirement prevents a claim of complete conformance.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements All Articles and Decisions 1–58; especially Decisions 25, 37–43, 47–51, 58. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.